Woxsen University – School of Technology
Object-Oriented Programming using Java
Project-Based Assignment (Replacement for Coursera Component)

Project Title: Smart Banking System (Console-Based Java Application)

Submission Deadline: 3–4 weeks before the External Examination (Hard Copy + Soft Submission Mandatory)

1. Objective
Design and implement a console-based banking management system using ObjectOriented Programming (OOP) principles in Java. The project should simulate real-world banking operations while demonstrating proper software design practices.

2. Functional Requirements
- Account Creation (Auto-generated account number, account holder name, account type, initial deposit).
- Deposit Operation (Validate deposit amount and update balance accordingly).
- Withdrawal Operation (Check minimum balance rule and handle insufficient balance conditions).
- Display Account Details (Account number, holder name, account type, current balance).
- Menu-Driven Interface (Create Account, Deposit, Withdraw, View Account, Exit).

3. Technical Requirements
- Use Classes and Objects appropriately.
- Use Constructors for object initialisation.
- Implement Encapsulation (private variables with getters/setters). • Implement Inheritance (SavingsAccount extends BankAccount).
- Use ArrayList to store and manage multiple accounts.
- Implement Exception Handling, including a custom InsufficientBalanceException.
- Follow proper naming conventions and maintain clean code structure.
- Ensure proper indentation and documentation within the code.

4. Submission Guidelines
- Submit all .java source files (properly named).

- Attach screenshots showing successful program execution.
- Include a short technical report (2–3 pages) explaining:
- - Class design and UML structure (if applicable).
- - OOP concepts implemented.
- - Challenges faced and how they were resolved.
- Hard copy submission in printed format.
- Soft copy submission on the ***shared folder*** clearly labelled with Name, Roll Number, and Section.

5. Assessment Rubric (Total: 100 Marks; Scale to 10%)

Component	Marks
Class Design & Structure	10
Implementation of OOP Concepts	20
Business Logic Correctness	20
Exception Handling (including custom exception)	5
Menu Implementation & User Interaction	10
Code Quality, Documentation & Naming Conventions	10
Innovation / Additional Features	5
Viva / Demonstration & Understanding	20

6. Academic Integrity

This project must be completed individually unless otherwise instructed. Viva voce will be conducted to verify individual understanding.

Issued By:
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